

SAHIL KUMAR SHARMA

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PROFESSIONAL SUMMARY

Software engineer with experience turning manual workflows into automated systems and building applications that help people collaborate and manage their work. Comfortable working across an application, from user-facing features to the services behind them, with attention to how the pieces work together.

EXPERIENCE

Tata Consultancy Services (TCS) | Cloud Application Engineer

December 2025 - Present

- Built an AI-powered outbound lead pipeline using Amazon Bedrock (Claude & Nova), Python-based AWS Lambda functions, S3 and a four-table DynamoDB state architecture, translating plain-English requests into executable filtering rules with validation and automated retries for 10K+ records per batch, alongside real-time DND checks and IST timezone normalization.
- Modernized legacy DTMF-based IVRs into conversational voice bots using Amazon Connect and Amazon Lex, automating customer self-service and call routing across 15K+ monthly interactions.
- Optimized 5+ event-driven backend services and API Gateway endpoints, reducing rule-processing latency to under two seconds and manual campaign operations effort by 45%.

Fundbridge Consultation Ltd. | SDE Intern

July 2025 - December 2025

- Implemented 13+ RESTful API endpoints using Node.js, Express and PostgreSQL, supporting investment and administration workflows with authenticated access and database-backed CRUD operations.
- Developed a full-stack product management platform, integrating secure authentication, media management and automated financial calculations into product workflows.
- Integrated payment gateway services and debugged production issues to improve transaction reliability and application stability.

PROJECTS

Smart Attendance System

[Live](#) [GitHub](#)

- Built a facial attendance system using ArcFace's 512-dimensional embeddings and cosine k-nearest-neighbor matching, incorporating enrollment quality checks, duplicate-face detection, multi-frame voting and MediaPipe-based blink challenges for active liveness.
- Developed FastAPI services with JWT authentication and two access roles, Admin and User, using PostgreSQL and SQLAlchemy to persist face embeddings, enrollment images and attendance records, with analytics for attendance trends, punctuality and anomalies.

Pulse

[Live](#) [GitHub](#)

- Developed a real-time Kanban application using Next.js, TypeScript, NestJS and PostgreSQL, supporting 3 task stages with drag-and-drop updates, assignments, comments and activity tracking.
- Implemented Socket.IO synchronization across isolated project rooms and 3 server-enforced access roles, Owner, Editor and Viewer, with JWT authentication, expiring invitations and connection cleanup on logout.

SKILLS

- Languages:** Python, Java, JavaScript, TypeScript, SQL, Bash
- Backend:** Node.js, Express.js, NestJS, FastAPI, Flask
- Frontend:** React, Next.js
- AWS:** Lambda, Bedrock, Connect, Lex, API Gateway, S3, DynamoDB, CloudFront, EC2, IAM
- Databases & ORMs:** PostgreSQL, MySQL, MongoDB, SQLite, Prisma, SQLAlchemy
- AI & Computer Vision:** InsightFace (ArcFace), MediaPipe, OpenCV, scikit-learn
- Tools & Technologies:** Git, GitHub Actions, Docker, Terraform, Socket.IO, JWT, OAuth

CERTIFICATIONS & ACHIEVEMENTS

- AWS Certified Cloud Practitioner CLF-C02** | 2026
- Top 1% Performer** – EY Cyber Security Boot Camp | 2024
- Selected for Amazon ML Summer School** | 2024
- AWS Academy Cloud Foundations, Architecting & Security** | 2024
- Azure Fundamentals Cloud Skills Challenge** | 2023

EDUCATION

ITER, Siksha 'O' Anusandhan University

B.Tech, Computer Science and Engineering

November 2021 - August 2025

Bhubaneswar, Odisha